a larger angle

5 **(ai)** $V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{170}{\sqrt{2}} = 120\text{ V}$ A1

(aii) $\omega = 2\pi/T = 314$

$T = 0.0200\text{ s}$ A1

(b) **(i)** $\frac{V_S}{V_P} = \frac{N_S}{N_P}$ C1

$\frac{V_S}{170} = \frac{3500}{2000}$

$V_S = 298\text{ V}$

A1

(ii)

$V_S = I_S R$ C1

$298 = I_S(130)$

$I_S = 2.288$ A

$\frac{I_P}{I_S} = \frac{N_S}{N_P} = \frac{3500}{2000}$ A1

$I_P = 4.00$ A

b **iii**

$V_S = I_S R$ C1

$298 = I_S(130)$

$I_S = 2.288$ A

$\frac{I_P}{I_S} = \frac{N_S}{N_P} = \frac{3500}{2000}$

$I_P = 4.00$ A

A1

Power loss due to B1
Induced Eddy currents
Hysteresis los
Any possible causes

(c)